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A lightweight transfer learning based ensemble approach for diabetic retinopathy detection

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ABSTRACT

Diabetic retinopathy (DR) is a fatal and irreversible eye disease that affects millions of people worldwide. It occurs due to high blood sugar level in the body of a diabetic patient, so it requires immediate attention which goes beyond the clinical solutions. With the advancements in deep learning and computer vision there are maximum possibilities of predicting this disease at early stages. Based on the severity of disease, different labels have been assigned to different classes of this disease as follows: 4 for proliferative DR, 3 for severe DR, 2 for moderate DR, 1 for mild DR and 0 for No DR. In this paper we proposed a deep learning-based ensemble approach using pre-trained and customized bi-class (CNN) base-learners like MobileNet, InceptionV3and DenseNet121 which were identified during initial investigation. These deep learning models were used as the base learner because of their promising performance in ensembles compared to the other deep learning base learners. All the work in the literature has studied this as a single complex multi-class problem or a bi-class problem where earlier stages are grouped together (0 to 3) and treated as one class and 4 as separate another class. Our work breaks this multi-class problem into multiple simpler two class problems using OVO(One-Versus-One) approach. Several benchmark data sets such as APTOS 2019, IDRiD, Messidor-2 and DDR which are multi-class data sets were used for training and testing our models. Data augmentation techniques were also utilized. Performance metrics such as precision, recall, f1-score, and accuracy were used for evaluation. Our ensemble models showed a remarkable performance with precision, recall, f1-score, and accuracy for most of the datasets used in this study. In addition to this our ensemble models have minimum number of trainable parameters which makes them an ultimate choice.

1. Introduction

Diabetic retinopathy (DR) is a condition that can cause vision loss and blindness in diabetic patients. It may affect patient's blood vessels in the retina. Diabetic retinopathy may not have any symptoms at earlier stages so it's very important for diabetic patients to get their eyes examined at least once a year. Detecting it early can help you take the necessary steps to save your vision. DR is classified in to two major categories: Proliferative Diabetic Retinopathy (PDR) and Non-Proliferative Diabetic Retinopathy (NPDR) (Memon et al, 2017). The DR in final stage is called Proliferative Diabetic Retinopathy (PDR) while as DR in earlier stages is referred to as Non-Proliferative Diabetic Retinopathy (NPDR). And this (NDPR) is further classified into Severe, Moderate and Mild stages. It's difficult to differentiate between Normal and Mild as they resemble each other closely. The sample of each stage is shown below in Fig. 1.

According to a report by WHO diabetes is increasing at a faster rate and it is expected to reach 700 million by 2045.Therefore keeping these facts and figures in mind the diagnosis and early screening of DR are of vital importance. Health professionals diagnose DR using fundus images which is a time-consuming process and requires expertise. Due to the shortage of experienced health professionals and the increasing number of DR patient's misdiagnosis is a common issue. Computerized diagnosis can greatly overcome this problem by reducing the workload of health professionals and thereby increasing the accuracy of diagnosis. In recent years deep learning has contributed a lot in the field of computerized diagnostics. Convolutional Neural Networks (CNN) is the most widely used and effective in the field of computer vision because of its performance in image classification. Multiple CNN algorithms like ResNet (He et al, 2016), GoogleNet (Szegedy et al, 2015), EfficientNet (Tan et al, 2019) and VGGNet (Simoyan et al, 2014) are of great significance. The revolutionized progress in the medical fields such as cancer

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